Problem

Stewart, *Calculus: Early Transcendentals*, Section 15.7 (Triple Integrals), Exercise 29.

Express the integral $\iiint_E f(x,y,z)\,dV$ as an iterated integral in six different ways, where E is the solid bounded by the given surfaces:

$$y = 4 - x^2 - 4z^2, \qquad y = 0.$$

Part (a)

Notation

The symbol E is the solid region in space. The symbol dV is the volume element. When the differentials are written in the order $dz\,dy\,dx$ the volume element is

$$dV = dz\,dy\,dx.$$

An *iterated integral* evaluates $\iiint_E f\,dV$ as three nested single integrals. Fubini's theorem states that the value does not depend on the order of integration when f is continuous on E and every order that fits the geometry of E is used. That theorem is why six different orders can all be correct.

Step 1: the two surfaces

The surface $y = 4 - x^2 - 4z^2$ is an elliptic paraboloid. Completing the square is not needed, because the equation already has the form

$$y = 4\left(1 - \left(\frac{x}{2}\right)^2 - z^2\right).$$

The surface opens toward negative y. Its vertex is the point $(0,4,0)$, where y is largest. A point (x,y,z) lies on the surface exactly when $y = 4 - x^2 - 4z^2$, so the *inside* of the paraboloid is $y \le 4 - x^2 - 4z^2$.

The second surface is the plane $y = 0$. These two surfaces meet where

$$4 - x^2 - 4z^2 = 0, \qquad \text{that is,} \qquad x^2 + 4z^2 = 4.$$

This is an ellipse in the plane $y = 0$. Its semi-axis is 2 along the x-axis and 1 along the z-axis. The ellipse is the rim of the solid, and E is the cap between the ellipse and the vertex:

$$E = \{(x,y,z) : x^2 + 4z^2 \le 4,\ 0 \le y \le 4 - x^2 - 4z^2\}.$$

The solid has one vertex, $(0,4,0)$, and one edge, the ellipse $x^2 + 4z^2 = 4$ in the plane $y = 0$. Figure 1 shows E.

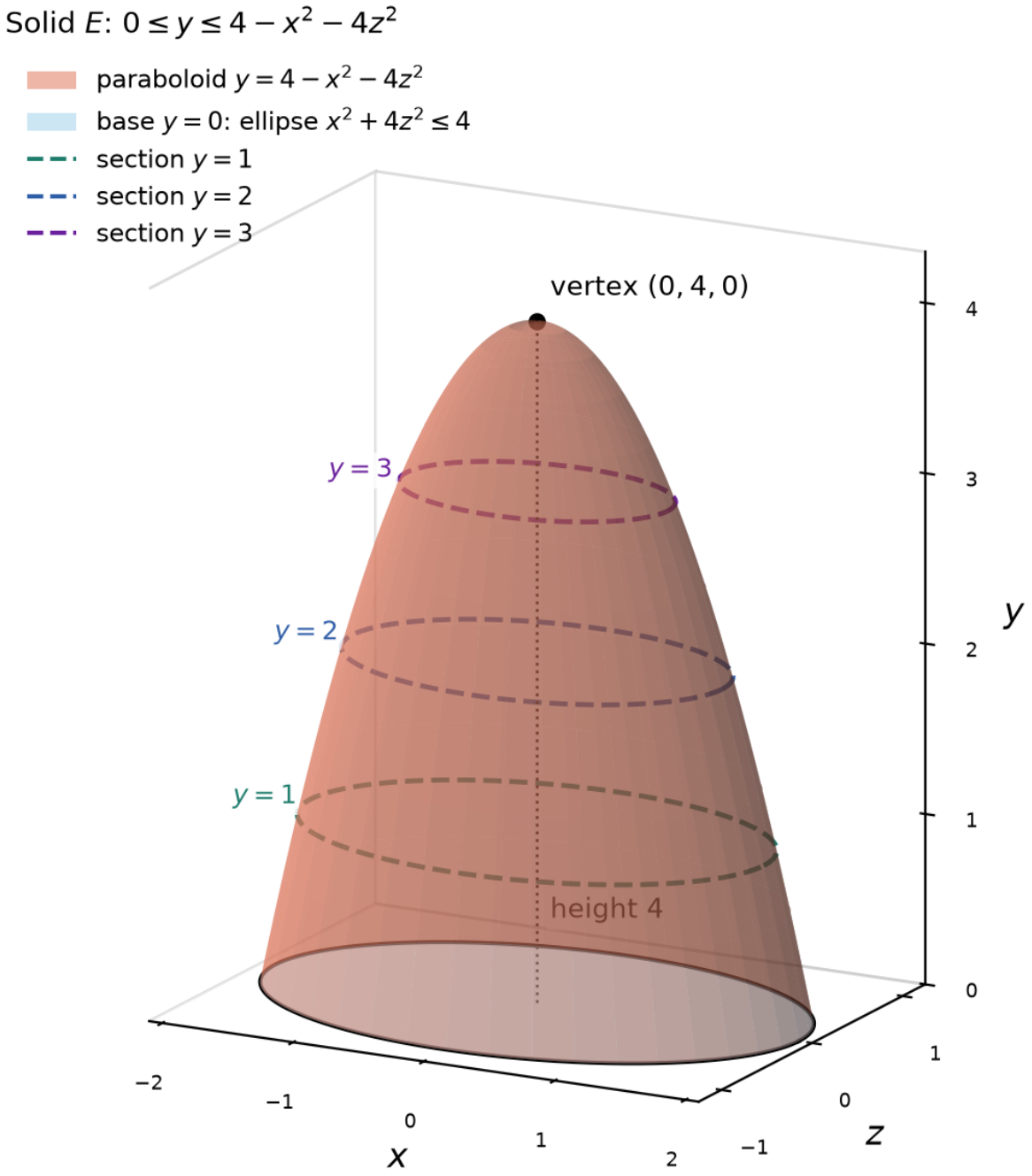


Figure 1: The solid E of Exercise 29. The floor is the ellipse $x^2 + 4z^2 \le 4$ in the plane $y = 0$. The roof is the elliptic paraboloid $y = 4 - x^2 - 4z^2$ with vertex $(0, 4, 0)$. The dashed curves are the sections $y = 1$, $y = 2$ and $y = 3$. The height of the solid is 4.

Step 2: the shadow of E

The shadow of E on the plane $y = 0$ is the set of (x, z) for which the vertical line over (x, z) meets E. That set is the ellipse

$$D = \{(x, z) : x^2 + 4z^2 \le 4\}.$$

The ellipse can be read in two ways, and each way gives a different pair of outer limits.

- Fix x with $-2 \le x \le 2$. Then $4z^2 \le 4 - x^2$, so $-\frac{1}{2}\sqrt{4 - x^2} \le z \le \frac{1}{2}\sqrt{4 - x^2}$.
- Fix z with $-1 \le z \le 1$. Then $x^2 \le 4 - 4z^2$, so $-2\sqrt{1 - z^2} \le x \le 2\sqrt{1 - z^2}$.

The vertical line over a point of D meets E in $0 \le y \le 4 - x^2 - 4z^2$, because the floor is $y = 0$ and the roof is the paraboloid. Figure 2 shows the two readings of D.

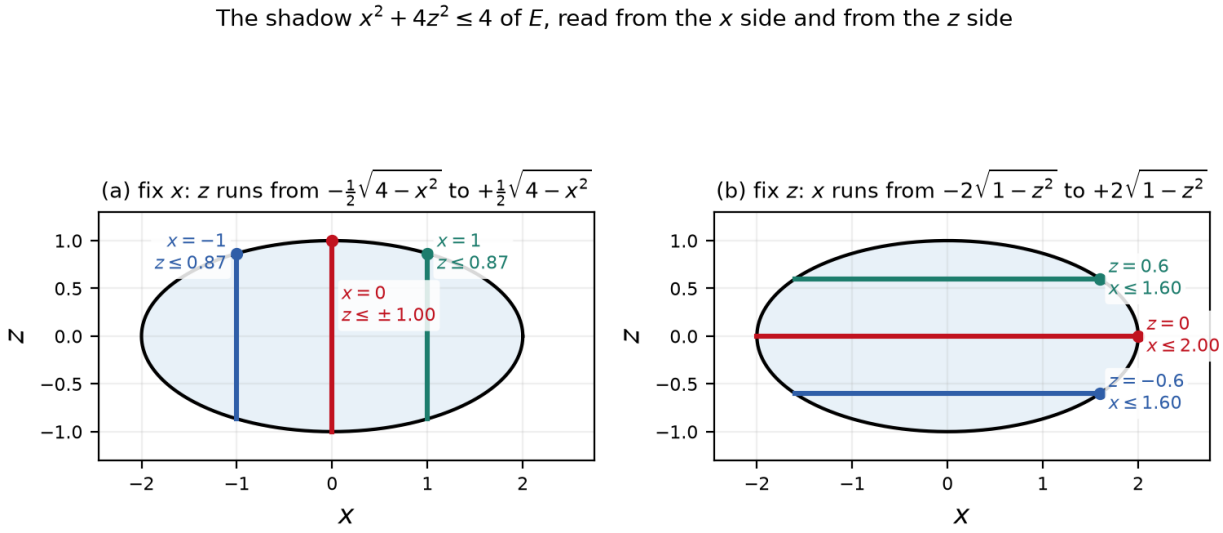


Figure 2: The shadow D of E in the plane $y = 0$ is the ellipse $x^2 + 4z^2 \le 4$. (a) With x fixed, z runs between the two halves of the ellipse. (b) With z fixed, x runs between the two halves. The two readings give the two possible pairs of outer limits.

Step 3: separate the variables

A pair of limits is legal when it contains only variables that are still free at that stage. In the order $dz\,dy\,dx$ the innermost pair is the z pair, so it may contain x and y. The middle pair is the y pair, so it may contain x only. The outer pair is the x pair, so it contains no variable.

Two different ways to reach the innermost variable are open. The first way uses the roof: after the x pair and the y pair are chosen, the variable z is still free, and it runs between the two halves of the ellipse $x^2 + 4z^2 = 4 - y$, whose height is $z = \pm\frac{1}{2}\sqrt{4 - y - x^2}$. The second way uses the floor: after the x pair and the z pair are chosen, the variable y runs from 0 to the roof, $y = 4 - x^2 - 4z^2$.

Step 4: the six iterated integrals

The six orders follow from these two choices. Table 1 lists them; each entry is $\iiint_E f(x,y,z)\,dV$ written in that order.

Table 1: The six orders of integration that fit the solid E of Exercise 29. In every row the inner limits may use the two outer variables, and the middle limits may use the outer variable.

Order	Inner limits	Middle limits	Outer limits
$dz\,dy\,dx$	$z:-\frac{1}{2}\sqrt{4-x^2-y} \to \frac{1}{2}\sqrt{4-x^2-y}$	$y:0\to 4-x^2$	$x:-2\to 2$
$dz\,dx\,dy$	$z:-\frac{1}{2}\sqrt{4-y-x^2} \to \frac{1}{2}\sqrt{4-y-x^2}$	$x:-\sqrt{4-y}\to\sqrt{4-y}$	$y:0\to 4$
$dy\,dz\,dx$	$y:0\to 4-x^2-4z^2$	$z:-\frac{1}{2}\sqrt{4-x^2}\to\frac{1}{2}\sqrt{4-x^2}$	$x:-2\to 2$
$dy\,dx\,dz$	$y:0\to 4-x^2-4z^2$	$x:-2\sqrt{1-z^2}\to 2\sqrt{1-z^2}$	$z:-1\to 1$
$dx\,dz\,dy$	$x:-\sqrt{4-y-4z^2} \to \sqrt{4-y-4z^2}$	$z:-\frac{1}{2}\sqrt{4-y}\to\frac{1}{2}\sqrt{4-y}$	$y:0\to 4$
$dx\,dy\,dz$	$x:-\sqrt{4-y-4z^2} \to \sqrt{4-y-4z^2}$	$y:0\to 4-4z^2$	$z:-1\to 1$

Written out, the six answers are

$$(1)\quad \int_{-2}^{2}\int_{0}^{4-x^2}\int_{-\frac{1}{2}\sqrt{4-x^2-y}}^{\frac{1}{2}\sqrt{4-x^2-y}} f(x,y,z)\,dz\,dy\,dx,$$

$$(2)\quad \int_{0}^{4}\int_{-\sqrt{4-y}}^{\sqrt{4-y}}\int_{-\frac{1}{2}\sqrt{4-y-x^2}}^{\frac{1}{2}\sqrt{4-y-x^2}} f(x,y,z)\,dz\,dx\,dy,$$

$$(3)\quad \int_{-2}^{2}\int_{-\frac{1}{2}\sqrt{4-x^2}}^{\frac{1}{2}\sqrt{4-x^2}}\int_{0}^{4-x^2-4z^2} f(x,y,z)\,dy\,dz\,dx,$$

$$(4)\quad \int_{-1}^{1}\int_{-2\sqrt{1-z^2}}^{2\sqrt{1-z^2}}\int_{0}^{4-x^2-4z^2} f(x,y,z)\,dy\,dx\,dz,$$

$$(5)\quad \int_{0}^{4}\int_{-\frac{1}{2}\sqrt{4-y}}^{\frac{1}{2}\sqrt{4-y}}\int_{-\sqrt{4-y-4z^2}}^{\sqrt{4-y-4z^2}} f(x,y,z)\,dx\,dz\,dy,$$

$$(6)\quad \int_{-1}^{1}\int_{0}^{4-4z^2}\int_{-\sqrt{4-y-4z^2}}^{\sqrt{4-y-4z^2}} f(x,y,z)\,dx\,dy\,dz.$$

Step 5: the volume, as a check

Set $f=1$. The volume is easiest to get from the sections at a fixed height y. At height y the section is the ellipse $x^2+4z^2\le 4-y$, whose semi-axes are $\sqrt{4-y}$ and $\frac{1}{2}\sqrt{4-y}$. Its area is

$$A(y)=\pi\cdot\sqrt{4-y}\cdot\frac{\sqrt{4-y}}{2}=\frac{\pi(4-y)}{2},$$

so

$$V=\int_0^4 A(y)\,dy=\int_0^4\frac{\pi(4-y)}{2}\,dy=\frac{\pi}{2}\left[4y-\frac{y^2}{2}\right]_0^4=\frac{\pi}{2}\,(16-8)=4\pi.$$

The same value follows from any of the six orders. For example, order (3) gives

$$V = \int_{-2}^{2} \int_{-\frac{1}{2}\sqrt{4-x^2}}^{\frac{1}{2}\sqrt{4-x^2}} \left(4 - x^2 - 4z^2\right) dz\, dx = \int_{-2}^{2} \tfrac{2}{3}\left(4 - x^2\right)^{3/2} dx = \tfrac{2}{3} \cdot 6\pi = 4\pi,$$

because $\int_{-a}^{a} (a^2 - x^2)^{3/2} dx = \frac{3\pi}{8} a^4 = 6\pi$ for $a = 2$. A third way is the substitution $u = x/2$, which turns the solid into a paraboloid of revolution $y = 4(1 - u^2 - z^2)$ and gives $V = 2 \int_{u^2+z^2 \le 1} 4(1 - u^2 - z^2)\, du\, dz = 4\pi$.

Table 2 reports the numerical values of the six orders, computed with Gauss–Legendre quadrature, for three integrands. All six orders agree, and all agree with the exact values from the definition of E.

Table 2: Numerical values of the six orders of Exercise 29 (Gauss–Legendre, 160^3 nodes).

Order	$f = 1$ (volume)	$f = y$	$f = x^2 + z^2$
$dz\, dy\, dx$	12.566370855	16.755162384	10.471975824
$dz\, dx\, dy$	12.566371432	16.755162944	10.471979763
$dy\, dz\, dx$	12.566370614	16.755160819	10.471975511
$dy\, dx\, dz$	12.566370614	16.755160819	10.471975512
$dx\, dz\, dy$	12.566371432	16.755162944	10.471976574
$dx\, dy\, dz$	12.566370855	16.755162384	10.471975590
exact	$4\pi = 12.566370614$	$16\pi/3 = 16.755160819$	$10\pi/3 = 10.471975512$

The largest difference between any two orders is $8.2 \cdot 10^{-7}$ for $f = 1$, $2.1 \cdot 10^{-6}$ for $f = y$ and $4.3 \cdot 10^{-6}$ for $f = x^2 + z^2$, which is the level of the quadrature error.

$$\boxed{E = \{(x, y, z) : x^2 + 4z^2 \le 4,\ 0 \le y \le 4 - x^2 - 4z^2\}, \qquad \iiint_E dV = 4\pi}$$

The six answers are the entries (1)–(6) above, with the limits collected in Table 1.

How to solve problems of this kind

1. Write the given surfaces as inequalities. The solid is the set of points that satisfy all of them. Here $y \geq 0$ and $y \leq 4 - x^2 - 4z^2$.
2. Find where two surfaces meet. The curve of intersection is the rim of the solid, and it fixes the outer limits of every order.
3. Draw the shadow of the solid on a coordinate plane. Then draw the shadow in two ways: as vertical strips with x fixed, and as horizontal strips with z fixed.
4. For each order, list the variables from the innermost differential outward. A limit may use only variables that are still free at that stage. This test rules out illegal orders immediately.
5. Choose the innermost variable in one of two ways: cut with a line parallel to the axis of that variable, or start from the flat face and stop at the curved face. The two choices give the two groups of three orders.
6. Check the limits with numbers: use $f = 1$, which must give the volume, and one non-constant function f. All orders must agree.